OLS Assumptions at a Glance: SLR.1–5, MLR.1–6

What each assumption buys, in finite and large samples — with the lecture example attached to each

#	Simple (SLR)	Multiple (MLR)	Role / exam point
1	Linear in parameters	Linear in parameters	Linear in coefficients; regressors may be logged or squared.
2	Random sampling	Random sampling	Sample speaks for the population. **OLS requires random sampling; it may also require random assignment.**
3	Sample variation in x	No perfect collinearity	Exact collinearity → no identification. **Dummy trap: session has 6 values → 5 dummies.**
4	$E(u\|x) = 0$	$E(u\|x_1,\ldots,x_k) = 0$	**Core of unbiasedness. Ability left in u and related to educ → biased, even with random sampling.** Consistency needs only MLR.4′.
5	$Var(u\|x) = \sigma^2$	$Var(u\|X) = \sigma^2$	Variance formula, **BLUE (Gauss–Markov)**, large-sample t/F.
6	—	$u\|X \sim Normal(0, \sigma^2)$	Exact t/F in any n; **not needed in large samples** (Ch. 5).

How to read the highlights

highlighted text = textbook point the professor stressed in class

★ 교수님 강조 card = lecture-only note (Econ 1–3, R labs)

1–4 → unbiased and consistent; 1–5 → BLUE and large-sample t/F; add MLR.6 → exact t/F in any sample.

Sources: Chapter 2, slides 28–36; Chapter 3, slides 18–22, 36–38; Chapter 4, slides 5–12; Chapter 5, slides 4–10; Econometrics 3, slides 2–5 · Lecture 10.01, 10.13

OLS Variance, Multicollinearity, Gauss–Markov

Three drivers of precision; collinearity costs variance, never bias

Theorem 3.2 — Slope variance (MLR.1–5)

$\mathrm{Var}(\beta_j|X) = \sigma^2 / [\mathrm{SST}_j(1 - R_j^2)]$

- $\sigma^2 \uparrow$ (more noise) → variance ↑
- **$\mathrm{SST}_j \uparrow$ (more spread in x_j) → variance ↓**; SST_j grows with n
- **$R_j^2 \uparrow$ (x_j explained by the other x's) → variance ↑**

Simple regression is the $R_j^2 = 0$ case: $\mathrm{Var}(\beta_1) = \sigma^2 / \mathrm{SST}_x$

Theorem 3.3 — Standard error

$\hat{\sigma}^2 = \mathrm{SSR}/(n - k - 1), \quad \mathrm{se}(\beta_j) = \hat{\sigma} / \sqrt{[\mathrm{SST}_j(1 - R_j^2)]}$

Valid only under homoskedasticity (MLR.5).

★ 교수님 강조 · Sept. lecture

If the variance of **sunlight** increases, the accuracy of $\hat{b}_1$ increases ($\mathrm{SST}_j \uparrow$).

If the explanatory variables are highly related, accuracy decreases ($R_j^2 \uparrow$) → **multicollinearity**.

Multicollinearity

$\mathrm{VIF}_j = 1 / (1 - R_j^2)$

Raises variance, **not bias**; not an MLR.3 violation (only perfect collinearity is). Only the variables involved are affected.

★ 교수님 강조 · 10.15 · Ch. 6 slide 23

Adding a variable that is **not (or only slightly) correlated** with the others but explains y lowers SSR → $\hat{\sigma}^2 \downarrow$ → **$\mathrm{se}(\beta_j) \downarrow$**, without adding collinearity.

Theorem 3.4 — Gauss–Markov

★ 교수님 강조 · Sept. lecture · "the most important theorem"

Under MLR.1–5, OLS is **BLUE**: smallest variance among linear unbiased estimators.

With heteroskedasticity OLS stays unbiased but is no longer best.

Collinearity inflates variance; omitted variables cause bias. Under MLR.1–5, OLS is **BLUE**.

Sources: Chapter 2, slides 31–36; Chapter 3, slides 27–38; Chapter 6, slide 23 · Lecture notes (Sept.), 10.15

Omitted-Variable Bias and Random Assignment

What is left in u decides whether β is unbiased

Omitted-variable bias — simple case

True: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$; estimated: y on x_1 only

$E(\beta_1) = \beta_1 + \beta_2 \delta_1 \quad \rightarrow \quad \text{bias} = \beta_2 \cdot \delta_1$

No bias if $\beta_2 = 0$ or x_1 and x_2 are uncorrelated.

	Corr(x_1, x_2) > 0	**Corr(x_1, x_2) < 0**
$\beta_2 > 0$	Positive bias	Negative bias
$\beta_2 < 0$	Negative bias	Positive bias

- Irrelevant variable included: still unbiased; may cost variance.
- Omitting one of several: generally biases **all** coefficients.

★ 교수님 강조 · Econometrics 3 · random assignment

Model 1: $\text{wage} = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{abil} + u$

Model 2: $\text{wage} = \beta_0 + \beta_1 \text{educ} + u$

In Model 2, ability goes into u. Abler people may get more education, and ability may raise wages → $E(u|\text{educ}) \neq 0$: **biased even with random sampling.**

Fix: add ability as a **control** (Model 1), or **randomly assign** educ so it cannot be related to u.

★ 교수님 강조 · 10.13 menu lab · interest became significant

Why? **Omitted-variable bias.** In Toussaert (2018) the coefficient on Interested rises 0.371 → 0.459 once beliefs are added: interest and beliefs correlate (+), beliefs lower WTP (−) → omitting beliefs biased interest **down**.

Larger estimate → larger |t| → **smaller p** (se also rose with collinearity, but less).

Bias = **$\beta_2 \cdot \delta_1$**: it needs a variable that both affects y and moves with x. **Controls or random assignment** remove it.

Sources: Chapter 3, slides 23–26, 41; Econometrics 3, slides 2–5; Toussaert (2018), Table VIII · Lecture 10.01, 10.13

Partial Slopes and Linear Relationships

β_1 in a multiple regression holds the other x's fixed; linearity is judged in the variables the model uses

Slope vs. partial slope

y on x_1 alone: β_1 is the slope. With other regressors, **β_1 is the partial slope**.

$\beta_1 = \beta_1 + \beta_2\delta_1$ (δ_1: slope of x_2 on x_1); equal only if $\beta_2 = 0$ or $\delta_1 = 0$

★ 교수님 강조 · 10.01 · reject in simple, fail to reject in multiple

Corr(skill, ravens) = 0.25 (VIF ≈ 1.06): se barely moves. The drop is mainly in the **estimate** — each simple slope also carried the other's effect.

social.csv	Estimate	se	p
skill alone	0.214	0.107	**0.049**
ravens alone	0.217	0.107	**0.046**
skill (both)	0.171	0.110	0.122
ravens (both)	0.175	0.110	0.115

Linear relationships (slides 7–12)

- Subsidy: linear with **log(price)**, not with price.
- **log(sunlight)** is linear with log(price); sunlight is not linear with price.

Sunlight × battery interaction (slides 13–15)

$$\log(\text{price}) = \ldots + \beta_3 \text{battery} + \beta_4(\text{sunlight} \times \text{battery}) + u$$

$$\partial \log(\text{price})/\partial \text{battery} = \beta_3 + \beta_4 \cdot \text{sunlight}$$

★ 교수님 강조 · Sept. lecture

Battery **has** a linear relationship with log(price) **if all others are held constant** — at a fixed sunlight the slope is one number.

Battery **does not** have a linear relationship where $\Delta\log(\text{price})/\Delta\text{battery}$ is **constant for all levels of sunlight** — the slope changes with sunlight.

Equations reconstructed from the slide text (the slide equations are images).

Partial slope = effect holding the other x's fixed. With an **interaction**, one variable's slope depends on the other.

Sources: Econometrics 2, slides 1–15; Chapter 3, slide 14; Chapter 6, slide 13; social.csv rerun of regression.R · Lecture (Sept.), 10.01

t Tests, p-Values, and “Fail to Reject”

Choose H_1 before seeing results; a large p-value is not proof of H_0

Alternative H_1 (H_0: $\beta_j = 0$)	Type	Rejection rule (level α, critical value c)	Slide example (5%)
$\beta_j > 0$	Right-tailed	$t > c$ (upper tail α)	df = 28: $t > 1.701$
$\beta_j < 0$	Left-tailed	$t < -c$ (lower tail α)	df = 18: $t < -1.734$
$\beta_j \neq 0$	Two-tailed	$\lvert t \rvert > c$ (α/2 in each tail)	$\lvert t \rvert > 2.06$

t statistic (MLR.1–6)

$t = (\beta_j - a_j) \,/\, se(\beta_j) \;\sim\; t(n-k-1)$

Large df, two-sided 10 / 5 / 1%: **1.645 / 1.96 / 2.576**.

p-value

Smallest significance level at which H_0 is rejected.

Two-sided: $p = P(|T| > |t|)$; $p \le \alpha \Leftrightarrow$ reject.

$P(|T| > 1.85) = 0.0718$ → reject at 10%, not at 5%.

★ 교수님 강조 · 10.08

Fail to reject ≠ H_0 is true. “Absence of evidence is not evidence of absence.”

Small n → se↑ → |t|↓ → fail to reject; a large n can reject the same effect.

★ 교수님 강조 · Simulation · 10,000 samples

True slope 0.2, n = 10: the share of “Reject” = **power**. With a true slope of 0, the share ≈ α (**Type I error**).

Reject when **$p \le \alpha$**. Failing to reject means the data are **not decisive**, not that $\beta_j = 0$.

Sources: Chapter 4, slides 5–28; simple regression demonstration.R · Lecture 10.08

Confidence Intervals and Significance

Intervals show precision; a significant coefficient still needs a magnitude check

Confidence interval

$\beta_j \pm c \cdot se(\beta_j)$, c = 97.5th percentile of t(n − k − 1) (≈ 1.96)

The bounds are random. In repeated samples, 95% of intervals built this way cover the true β_j.

a_j outside the 95% CI ⇔ reject H_0: $\beta_j = a_j$ (5%, two-sided).

Large df: 90% → 1.645, 95% → 1.96, 99% → 2.576.

★ 교수님 강조 · 10.08 · ggplot of 20 intervals

Why are the interval lengths different? Each sample gives a different $\hat{\sigma}$ and SST_x, so **se(β₁) is random** — and so is the length 2c · se.

About 1 in 20 intervals misses the true slope 2.

Statistical vs. economic significance

- Significant → judge magnitude and units.
- Significant ≠ important: large n makes tiny effects significant.
- Important but wrong sign → suspect misspecification.

Testing a linear combination (H_0: $\beta_1 = \beta_2$)

$t = (\beta_1 - \beta_2) / se(\beta_1 - \beta_2)$ — se needs Cov(β₁, β₂), which is not in standard output.

Reparameterize $\theta_1 = \beta_1 - \beta_2$: regress on jc and totcoll = jc + univ → the coefficient on jc is θ_1, with its se.

A 95% CI covers the truth in **95% of repeated samples**; its width varies because **se is estimated**.

Sources: Chapter 4, slides 28–35; simple regression demonstration.R · Lecture 10.08

F Tests: Restricted vs. Unrestricted

Does imposing H_0 raise SSR by more than chance would?

Exclusion restrictions

H_0: $\beta_{k-q+1} = \ldots = \beta_k = 0$ (q restrictions)

$F = [(SSR_r - SSR_{ur})/q] \;/\; [SSR_{ur}/(n-k-1)] \;\sim\; F(q, n-k-1)$

★ 교수님 강조 · Slides 38–39 · 10.01

Moving from unrestricted to restricted can only **raise SSR** → F ≥ 0.

H_0 **false** → dropping real effects raises SSR a lot → **F is large**.

H_0 **true** → F lands in the upper 5% tail only 5% of the time → reject if F > c (p < α).

Overall: $F = (R^2/k) \;/\; [(1-R^2)/(n-k-1)]$. With q = 1, **F = t²**. R^2 form only with the same y.

★ 교수님 강조 · 10.01 · individually insignificant, jointly significant

social.csv: skill p = 0.12, ravens p = 0.12, yet **F(2, 82) = 3.3, p = 0.042** → at least one coefficient ≠ 0.

★ 교수님 강조 · 10.13 · individually significant, jointly insignificant

int alone: **reject**. tim alone: fail to reject. int and tim jointly: **fail to reject**.

"Is the int coefficient 0?" → No. "Are int and tim both 0?" → Maybe.

Pooling a real effect with a null one **dilutes** the F test — a limit of the testing method.

Use F, not repeated t tests, for joint hypotheses.

F compares **SSR with and without the restrictions**; joint and individual tests can disagree in **both** directions.

Sources: Chapter 4, slides 36–43; regression.R (social.csv); menu lab · Lecture 10.01, 10.13

Unbiasedness, Consistency, Asymptotic Normality

What OLS guarantees as n grows — without normal errors, but still with MLR.1–5

Property	Meaning	Needs
Unbiased	$E(\beta_j) = \beta_j$ at every n	MLR.1–4
Consistent	$\text{plim}\ \beta_j = \beta_j$ as $n \to \infty$	MLR.1–3 + MLR.4′

MLR.4′: $E(u) = 0$, $\text{Cov}(x_j, u) = 0$ — weaker than MLR.4.

★ 교수님 강조 · 10.15

Consistency: as n increases, the distribution of β_j collapses onto β_j — **the graph gets sharper**.

Biased but consistent: bias vanishes as n grows. Unbiased but inconsistent: e.g., using only the first observation.

Asymptotic omitted-variable bias

$\text{plim}\ \tilde{\beta}_1 = \beta_1 + \beta_2\delta_1$ — **does not shrink as n grows**.

Theorem 5.2 — Asymptotic normality (MLR.1–5)

$(\beta_j - \beta_j) / \text{se}(\beta_j) \ \to^a\ \text{Normal}(0, 1)$

The t statistic and the t distribution → standard normal: t tests, CIs and F tests stay valid without MLR.6.

MLR.1–5 are still needed — especially homoskedasticity.

Why large samples are better

$\text{se}(\beta_j) \approx c_j / \sqrt{n}$ — quadrupling n roughly halves se.

★ 교수님 강조 · 10.15 · limits in practice

No infinite sample exists — but the trend is visible as n grows. Samples are expensive.

MLR.4′ → consistency; MLR.1–5 → large-sample t/F without normality. Omitted-variable inconsistency never fades.

Sources: Chapter 5, slides 4–12 · Lecture 10.15

Logs, Quadratics, and Interactions

Rescaling changes coefficients but not significance; logs change interpretation

Model	y	x	Interpretation of β_1
Level–level	y	x	$\Delta y = \beta_1 \Delta x$
Log–level	log(y)	x	$\%\Delta y \approx (100\beta_1)\Delta x$
Level–log	y	log(x)	$\Delta y \approx (\beta_1/100)\%\Delta x$
Log–log	log(y)	log(x)	$\%\Delta y \approx \beta_1 \%\Delta x$ (elasticity)

★ 교수님 강조 · 10.15 · when to use / not use log

Benefits: % or elasticity reading; slopes invariant to rescaling; tames outliers; helps normality and homoskedasticity.

Harm / limits: do not log years, percentage points, or zero/negative values; hard to undo for prediction.

Units

$y \times c$: coefficients and se × c. $x_j \times c$: β_j and se ÷ c. **t and R^2 unchanged.**

Quadratic

$\Delta\hat{y} \approx (\beta_1 + 2\beta_2 x)\Delta x, \quad x^* = |\beta_1 \,/\, (2\beta_2)|$

Wage peaks near 24.4 years of experience — check how many observations lie beyond x*.

Interaction

$\partial y/\partial x_2 = \beta_2 + \beta_3 x_1$: β_2 is the effect at $x_1 = 0$, often meaningless → center: $\beta_3(x_1 - \mu_1)(x_2 - \mu_2)$.

APE: compute each unit's partial effect, then average over the sample.

Rescaling leaves **t and R^2 unchanged**; with quadratics or interactions the partial effect **depends on x**.

Sources: Chapter 2, slides 22–26; Chapter 6, slides 2–15 · Lecture 10.15

Choosing Regressors, Prediction, and R Essentials

R^2 is not the goal; predicting the mean differs from predicting one outcome

Adjusted R^2

$\bar{R}^2 = 1 - (1 - R^2)(n - 1)/(n - k - 1)$ — rises only if the new regressor has $|t| > 1$.

Never compare R^2 across different y's.

Too many controls

Do not control for the channel of the effect (beer consumption when regressing fatalities on beer taxes).

Prediction

Prediction interval is wider than the CI for the mean: $\text{se}(\hat{e}^0) = \sqrt{[\text{se}(\hat{y}^0)^2 + \hat{\sigma}^2]}$.

log(y) model: $\hat{y} = \hat{\alpha}_0 \cdot \exp(\log \hat{y})$, $\hat{\alpha}_0 = n^{-1} \Sigma \exp(\hat{u}_i)$.

★ 교수님 강조 · R lab essentials · 10.01 – 10.15

Code	What it does
`read.csv("social.csv", header=T)`	Load data
`data$new <- …`	Add a column
`data[ , 4]`	4th column (`data(,4)` is wrong)
`colnames(data)`	Column names
`lm(y ~ x1 + x2); summary(); confint()`	Fit, t/F/p, CI
`as.factor(session)`	Dummies, one base level dropped
`x1*x2`	x1 + x2 + x1:x2
`|  &  &&`	or · and (vectors) · and (single value)

Use **$\bar{R}^2$ only with the same y**; control for what the question requires; a prediction interval adds **$\hat{\sigma}^2$**.

Sources: Chapter 6, slides 16–25; regression.R, Lab4.R, simple regression demonstration.R · Lecture 10.01, 10.13, 10.15